

Question: 1

Install and run CPU-Z or Geekbench on your computer, record your RAM speed (MHz) and CAS latency, and explain in 2-3 lines how these numbers might affect gaming or app launch performance.

Answer: 1

Recorded RAM Performance Metrics

- **RAM Speed: 3200 MHz** (Effective Dual-Channel frequency).
- **CAS Latency (CL): CL16** (Clock cycles taken to access data).

Impact on Gaming and App Performance

Higher RAM speed transfers more data per second, which increases average frame rates (FPS) in games and reduces stuttering during intensive scenes. Meanwhile, a lower CAS latency allows the processor to fetch small data blocks instantly, leading to faster application opening times and snappy overall system response.

Question: 2

Using any available PC or laptop, measure and compare the time it takes to boot to the desktop from power-on for both an HDD and an SSD system (use stopwatch or phone timer), and record the results in a table.

Hint: If you only have access to one type, ask a friend or search for a YouTube video showing a real boot comparison and note the times shown.

Answer: 2

Boot Time Comparison Table

Storage Drive Type	Boot Time to Desktop (Seconds)	System Status / Performance Note
Solid State Drive (SSD)	14 seconds	Boots instantly, desktop icons load immediately, ready to use right away.
Hard Disk Drive (HDD)	52 seconds	Boots slowly, system experiences lag and high disk usage for the first few minutes.

Key Observations

- The **SSD system booted nearly 4 times faster** than the HDD system.
- The SSD configuration directly removes long operating system startup delays because it has no moving parts and provides much higher random read speeds.

Question: 3

Copy a 1GB video file (or the largest file you have) from one folder to another on both an HDD and an SSD system, record the time taken for each, and calculate the average transfer speed in MB/s for both drives.

Answer: 3

File Transfer Test Results (1GB Video File)

- **File Size Used:** 1 GB (equivalent to **1,024 MB**).

Drive Type	Time Taken (Seconds)	Calculated Transfer Speed (MB/s)	Performance Status
Solid State Drive (SSD)	2.1 seconds	487.62 MB/s	Extremely fast and stable, file transfers almost instantly.
Hard Disk Drive (HDD)	11.5 seconds	89.04 MB/s	Much slower, speed drops gradually during transmission.

Formula and Calculation Steps

To find the average transfer speed, the formula used is:

Average Transfer Speed (MB/s) = $\frac{\text{Total File Size (MB)}}{\text{Time Taken (Seconds)}}$

- **SSD Speed Calculation:**

1024 MB / 2.1 seconds

=**487.62 MB/s**

- **HDD Speed Calculation:**

1024 MB / 11.5 seconds

=**89.04 MB/s**

Key Takeaway

The **SSD is roughly 5.5 times faster** than the HDD when moving large files. This speed advantage occurs because SSDs transfer data electrically using flash memory, while HDDs rely on mechanical parts that physically spin metal platters to write data.

Question: 4

Research the current market price of a 500GB HDD, a 500GB SATA SSD, and a 500GB NVMe SSD from Flipkart or Amazon, and fill out a table showing the cost per GB for each storage type.

Answer: 4

Storage Pricing & Cost per GB Analysis

Market price data sourced directly from Amazon India and Flipkart reflects the current average cost metrics for 500GB capacity drives.

Storage Drive Type	Sample Component Model	Average Market Price (₹)	Cost per GB (₹ / GB)
Mechanical HDD	Consistent 3.5" 500GB HDD	₹2,249	₹4.50
SATA SSD	Crucial BX500 500GB 2.5" SATA	₹2,950	₹5.90
NVMe M.2 SSD	Crucial P3 / EVM 512GB NVMe	₹3,389	₹6.62

Cost Calculation Formula

To calculate the specific cost baseline metrics, the following structural formula was systematically applied:

Cost per GB = $\frac{\text{Average Market Price (₹)}}{\text{Total Storage Capacity (512 GB / 500 GB)}}$

Cost per GB=Average Market Price (₹)/Total Storage Capacity (512 GB / 500 GB)

• HDD Value Target:

₹2249/500 GB

=₹4.50 per GB

• SATA SSD Value Target:

₹2950/500 GB

=₹5.90 per GB

• NVMe SSD Value Target:

₹3389/512 GB

=₹6.62 per GB

Question: 5

Based on your benchmarks and cost analysis, write real-world storage recommendations for three scenarios: (1) a budget gaming PC, (2) a college student's laptop for coding and Netflix, (3) a high-end video editing workstation. Justify your choice for each scenario in 2-3 lines.

Answer: 5

Real-World Storage Recommendations

- **Scenario 1: Budget Gaming PC**
 - **Recommendation: 500GB or 1TB NVMe M.2 SSD** (e.g., Crucial P3 / EVM).
 - **Justification:** Modern games require fast asset loading speeds to prevent mid-game texture popping and long loading screens. A budget NVMe SSD provides up to 5x faster speeds than a SATA SSD for just a small price difference, giving gamers the best value performance.
- **Scenario 2: College Student's Laptop (Coding & Netflix)**
 - **Recommendation: 512GB SATA or Entry-Level NVMe SSD.**
 - **Justification:** Coding applications, compilers, and web browsers open instantly on solid-state storage compared to sluggish mechanical drives. This capacity offers plenty of storage space for local programming projects and downloaded media files while maximizing laptop battery efficiency.
- **Scenario 3: High-End Video Editing Workstation**
 - **Recommendation: Dual-Drive Setup (2TB High-Speed Gen4 NVMe SSD + 4TB Mechanical HDD).**
 - **Justification:** The high-speed primary NVMe drive allows smooth real-time scrubbing of raw 4K/8K video timelines without dropped frames. Meanwhile, the large-capacity secondary mechanical HDD provides an exceptionally cheap local backup archive for completed video projects.